# PIKVAIDYA AI (□□□□□□□□ AI)

Master Development & Operational Lifecycle Roadmap
Complete Architectural Blueprint & Step-by-Step Execution Plan across All 23 Phases

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| **Document UID:** | PV-ROADMAP-2026-v3.0 | **Current Baseline:** | Phase 16 Complete (240 / 240 Tests Passing, 100%) |
| **Date of Issuance:** | November 14, 2026 | **Target Horizon:** | Commercial Multi-State Field Deployment (Phases 17–23) |
| **Classification:** | System Architecture Specification | **Focus Region:** | Maharashtra Agrarian Ecosystem & All-India Agro-Climates |

Executive Overview: PikVaidya AI is an enterprise-grade multimodal artificial intelligence platform designed for smallholder farmers across India. It integrates calibrated convolutional neural networks, visual Grad-CAM explainability, real-time microclimate spray gating, trilingual Indic Hybrid RAG, in-situ IoT telemetry ingestion, and offline PWA edge diagnostics. This document delineates the exhaustive 23-phase lifecycle required to transition the project from its current verified backend foundation into a fully certified commercial field deployment.

## 1. Multi-Agent Engineering Governance Framework

In strict accordance with system engineering protocols (RULE[user_global]), the platform lifecycle is divided into 6 Core Engineering Divisions. Each division is supervised by an autonomous Monitoring Sentinel Agent and executed by a Lead Professional Software Engineer to guarantee modular isolation, eliminate cross-subsystem regressions, and enforce empirical test verification:

| Division & Scope | Monitoring Sentinel | Assigned Lead Engineer | Operational Responsibilities |
|---|---|---|---|
| Div 1: Deep Vision & XAI Phases 2, 3, 4, 5, 10, 17 | **Sentinel-VisionAI** | Lead Computer Vision & Deep Learning Engineer | Neural backbones (MobileNetV3, EfficientNet), Grad-CAM hooks, Platt calibration, ExG foliar masking, INT8 quantization. |
| Div 2: API Gateway & DB Phases 1, 6, 7, 15 | **Sentinel-Gateway** | Senior Backend Architect & Database Administrator | FastAPI async gateway, SQLAlchemy 2.0 models, Alembic migrations, rate limiting, and PBKDF2/JWT security. |
| Div 3: Microclimate & IoT Phases 9, 11, 19, 20 | **Sentinel-EdgeIoT** | Senior Distributed Systems & Embedded Engineer | LoRaWAN gateways, MQTT brokers, in-situ sensor telemetry, Open-Meteo models, spray window gating. |
| Div 4: Agronomic RAG Phases 8, 14, 18, 22 | **Sentinel-AgronomyRAG** | Principal Agronomy & NLP Systems Specialist | Trilingual BM25/Dense RRF retrieval, CIBRC chemical dosage compliance, statutory PHI/PPE rules, Bayesian fusion. |
| Div 5: Edge Client & PWA Phases 12, 13, 21 | **Sentinel-ClientEdge** | Senior Frontend & Mobile Applications Engineer | Service Worker precaching, 4-Store IndexedDB, ONNX WASM SIMD execution, ChatGPT conversational UI. |
| Div 6: Cloud Infra & QA Phases 16, 23 | **Sentinel-DevOpsQA** | Principal DevOps & Site Reliability Engineer | 100% pytest pass rate enforcement, Docker multi-stage builds, Prometheus telemetry, Kubernetes cluster scaling. |

## 2. Master Phase Execution Matrix (Phases 1 to 23)

The table below enumerates all 23 development phases. Phases 1 to 16 are Completed and Empirically Verified across 240 automated test suites. Phases 17 to 23 constitute the active roadmap toward commercial field deployment:

| Phase | Feature Name | Division | Status | Evidence / Key Deliverable |
|---|---|---|---|---|
| P-01 | Monorepo Topology & Environment | Div 2 | **VERIFIED** | Clean isolated packaging in pyproject.toml; standard structure. |
| P-02 | Vision Models (Sugarcane, Tomato, Onion) | Div 1 | **VERIFIED** | Trained checkpoints; Onion COLD 98.62% test acc; 4 models / 19 classes. |
| P-03 | Explainable AI (Grad-CAM Saliency) | Div 1 | **VERIFIED** | Target conv-layer hooks with normalized JET overlays; 7 tests pass. |

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| P-04 | Platt Temperature Scaling Calibration | Div 1 | **VERIFIED** | Optimized $T^*$ minimizing NLL; Expected Calibration Error < 0.05. |
| P-05 | Foliar ExG Masking & Blur Rejection | Div 1 | **VERIFIED** | Excess Green index + Otsu + Laplacian blur filter ($ ext{Var} < 65.0$). |
| P-06 | FastAPI Core Gateway & Singletons | Div 2 | **VERIFIED** | Async server, singleton memory weights, magic-byte MIME validation. |
| P-07 | SQLAlchemy Persistence & Audit Logs | Div 2 | **VERIFIED** | 10 declarative models, Alembic migrations, foreign key cascading. |
| P-08 | Trilingual Indic Hybrid RAG Assistant | Div 4 | **VERIFIED** | BM25 + Dense RRF ($C=60$), MR/HI/EN support, CIBRC/ICAR compliance. |
| P-09 | Hyperlocal Weather & Spray Gating | Div 3 | **VERIFIED** | Open-Meteo & IMD alerts for 11 districts; wind/rain spray holds. |
| P-10 | Unified Crop Health & Crop Gate | Div 1 | **VERIFIED** | MobileNetV3 Crop Gate (99.67% acc); unified /predict route. |
| P-11 | In-Situ IoT Telemetry Gateway | Div 3 | **VERIFIED** | LoRaWAN TTN/ChirpStack webhook, SHA-256 auth, in-situ RAG fusion. |
| P-12 | Offline PWA & INT8 WASM SIMD | Div 5 | **VERIFIED** | W3C PWA, 4-store IndexedDB, ONNX Web SIMD edge inference (2.3ms). |
| P-13 | ChatGPT Conversational Interface | Div 5 | **VERIFIED** | Unified chat canvas, tri-tool orchestrator, streaming SSE responses. |
| P-14 | Field Advisory PDF Report Generator | Div 4 | **VERIFIED** | ReportLab 4.4.4 legal-grade A4 advisory with side-by-side Grad-CAM. |
| P-15 | Human Agronomist Review & Triage | Div 2 | **VERIFIED** | ExpertReview state machine, triage queue, agronomist override. |
| P-16 | Offline Edge Sync & 19-Step E2E Test | Div 6 | **VERIFIED** | Bi-directional sync reconciliation; 19-step lifecycle test passed. |
| P-17 | Cotton & Soybean Crop Expansion | Div 1 | **IMMEDIATE** | ICAR-CICR/IISR real field datasets, MobileNetV3 fine-tuning, CIBRC seeds. |
| P-18 | Streaming Indic Voice (ASR & TTS) | Div 4 | **UPCOMING** | Whisper-small rural Marathi/Hindi ASR + FastSpeech2 neural TTS stream. |
| P-19 | Standalone Async MQTT Broker Daemon | Div 3 | **UPCOMING** | Direct MQTT queue subscriber (paho-mqtt) for solar field weather stations. |
| P-20 | Satellite Remote Sensing (Sentinel-2) | Div 3 | **UPCOMING** | 10m multispectral NDVI/NDWI foliar vigor & water stress polygon tracking. |
| P-21 | Mobile App Store Native Packaging | Div 5 | **UPCOMING** | Capacitor Android APK build with embedded ONNX WASM & camera macro lock. |
| P-22 | Multi-KVK Clinical Agronomic Trials | Div 4 | **UPCOMING** | Double-blind trials across 4 Maharashtra KVKs; target >95% concordance. |
| P-23 | Production Kubernetes Cloud Cluster | Div 6 | **UPCOMING** | Multi-region K8s deployment, PgBouncer pool, Redis cluster, Prometheus CI/CD. |

# 3. Technical Specifications of Completed Phases (Phases 1 to 16)

Every completed milestone represents an empirically validated subsystem backed by unit, integration, and end-to-end regression tests in the repository:

### Part I: Foundations & Deep Vision Engine (Phases 1 to 6)

• Phase 1 (Monorepo Topology): Standardized structure separating backend/, frontend/, models/, and data/ with unified packaging in pyproject.toml.
• Phase 2 (Deep Vision Models): 4 trained PyTorch models across 19 classes: Sugarcane EfficientNet-B0 (94.74%) and MobileNetV3 (93.68%), Tomato MobileNetV3 (96.00%), and Onion MobileNetV3 retrained on 5,317 COLD field leaves (98.62% test acc, 0.9885 macro F1).
• Phase 3 (Grad-CAM XAI): Saliency maps computed via final conv-layer gradient backpropagation ($L_{ ext{Grad-CAM}}$), normalized and overlaid onto $224 \times 224$ leaves with clean memory hook teardown.
• Phase 4 (Platt Temperature Scaling): Dynamic validation calibration ($T^* = 1.258$ Sugarcane, $0.835$ Tomato, $1.305$ Onion) cutting Expected Calibration Error (ECE) by up to 49.95%.
• Phase 5 (ExG Segmentation & Blur Guard): Excess Green ($2G - R - B$) with Otsu thresholding rejects non-foliar backgrounds (< 8% foliage). 2D discrete Laplacian filter rejects motion-blurred captures ($ ext{Var} < 65.0$).
• Phase 6 (Serving Gateway): FastAPI async engine with singleton model memory caching, magic-byte file signature sniffing, and SlowAPI rate limiting.

### Part II: Data Persistence, Agronomy & Microclimate (Phases 7 to 10)

• Phase 7 (Relational Persistence): SQLAlchemy 2.0 schema defining 10 entities (User, Farm, CropCycle, Diagnosis, WeatherObservation, IoTDevice, IoTTelemetry, Notification, ExpertReview, AuditLog) with Alembic migrations.
• Phase 8 (Indic Hybrid RAG): Hybrid BM25 lexical + dense semantic vector retrieval fused via Reciprocal Rank Fusion ($C=60$), cross-score concordance reranking, and CIBRC chemical prescriptions with strict PHI days and PPE warnings.
• Phase 9 (Microclimate Intelligence): Open-Meteo & IMD alert monitoring across 11 Maharashtra districts with spray suitability gating (winds $\ge 15$ km/h or precipitation $> 40\%$ automatically trigger chemical application holds).
• Phase 10 (Unified Crop Health Platform): MobileNetV3 Crop Detector with 99.67% test accuracy across 300 unseen leaves for automated crop routing, executing diagnosis, calibration, and microclimate risk under POST /api/crop-health/predict.

### Part III: Edge Computing, Offline PWA & IoT Telemetry (Phases 11 to 13)

• Phase 11 (In-Situ IoT Gateway): LoRaWAN webhook integration (TTN / ChirpStack) and authenticated REST endpoints (SHA-256 tokens) ingesting canopy temperature, leaf wetness minutes, and root soil moisture. Seamlessly integrated into the multimodal RAG orchestrator.
• Phase 12 (Offline PWA & INT8 WASM): Progressive Web App with custom service worker precaching 14 critical assets (118.3 MB), 4-store IndexedDB architecture, and ONNX Runtime Web WASM SIMD execution delivering 2.35 ms to 2.66 ms client-side diagnostic latency.
• Phase 13 (Conversational Multi-Tool UI): Central ChatGPT-style conversational canvas supporting text, multi-image, PDF, and voice inputs, orchestrating tri-tool execution (Diagnosis, Weather Risk, Indic Knowledge) via Server-Sent Events (SSE).

### Part IV: Enterprise Workflows, Reporting & Verification (Phases 14 to 16)

• Phase 14 (Field Advisory PDF Generator): ReportLab 4.4.4 legal-grade A4 advisory report generator featuring side-by-side visual leaf/Grad-CAM panels, spray safety banners, dosage tables, and statutory notices under the Insecticides Act, 1968.
• Phase 15 (Human Agronomist Review): ExpertReview triage state machine allowing certified extension officers to inspect ambiguous foliar scans and override AI diagnoses.
• Phase 16 (Edge Sync Reconciliation & E2E Validation): Offline sync push/pull endpoints with conflict resolution, corroborated by an automated 19-step end-to-end integration test validating the entire farmer lifecycle (240/240 tests passing).

# 4. Implementation Blueprints for Upcoming Phases (Phases 17 to 23)

To advance PikVaidya AI to full commercial field deployment, the following 7 phases will be executed in sequential sprints:

### Phase 17: Multi-Crop Expansion — Cotton & Soybean Vision Pipelines (Immediate Next)

• Target Crops: Cotton (Gossypium hirsutum) and Soybean (Glycine max) across Vidarbha and Marathwada.
• Pathological Classes: Cotton Bacterial Blight, Grey Mildew, Alternaria Leaf Spot, Healthy; Soybean Rust, Frogeye Spot, Yellow Mosaic Virus, Healthy.
• Execution Protocol: (1) Ingest certified real field images from ICAR-CICR and ICAR-IISR. (2) Apply parent-grouped SHA-256 stratification (70/15/15) to prevent augmented data leakage. (3) Fine-tune MobileNetV3-Large backbones with inverted residual block freezing and class-weighted focal loss. (4) Optimize temperature scalar $T^*$ for $ ext{ECE} < 0.05$ and quantize to INT8 ONNX.
• Acceptance Criteria: Test accuracy $\ge 95.0\%$, macro F1 $\ge 0.940$, INT8 CPU inference latency $< 25$ ms.

### Phase 18: Streaming Indic Voice Engine (Real-Time Marathi & Hindi ASR/TTS)

• Strategic Need: Enables hands-free, illiterate-friendly agricultural diagnosis in rural fields.
• Architecture: Streaming Whisper-small / Conformer ASR model fine-tuned on rural Marathi/Hindi dialects, paired with FastSpeech2/Kokoro-Indic neural TTS. Bi-directional WebSocket endpoint (WS /api/v1/chat/voice-stream).
• Acceptance Criteria: Full-duplex audio latency $< 850$ ms; Word Error Rate (WER) $< 12\%$ on noisy rural field audio.

### Phase 19: Standalone Asynchronous MQTT Broker Subscriber Daemon

• Strategic Need: High-frequency streaming telemetry ingest from remote solar-powered IoT weather nodes.
• Architecture: Asynchronous Python daemon using asyncio-mqtt subscribing to pikvaidya/farms/+/sensors/#. Implements QoS 1 persistence, automatic backpressure handling, and sliding-window fungal infection risk updates.
• Acceptance Criteria: Throughput $\ge 500$ messages/sec with zero dropped packets and $< 50$ ms batch persistence latency.

### Phase 20: Satellite Remote Sensing Integration (Sentinel-2 10m NDVI / Water Stress)

• Strategic Need: Macroscopic field-wide validation of ground foliar leaf scans using space-borne multispectral data.
• Architecture: Automated Copernicus Open Access Hub ingestion clipping 10m Sentinel-2 MSI bands over farmer cadastral polygons. Computes time-series NDVI (vigor), NDRE (chlorophyll), and NDWI (foliar water deficit) to detect early blight spreads.
• Acceptance Criteria: Automated 5-day revisit ingestion; polygon raster clipping $< 3$ seconds per registered estate.

### Phase 21: Mobile App Store Native Packaging (Capacitor / Android APK)

• Strategic Need: Google Play Store distribution as a standalone offline-first Android application.
• Architecture: Capacitor native runtime with hardware camera bindings (macro autofocus lock, auto-LED torch for foliar scans). Bundles INT8 ONNX models and WASM SIMD runtimes directly in the APK package to eliminate runtime model downloads.
• Acceptance Criteria: Release APK size $< 75$ MB; executes full offline diagnosis in airplane mode on budget Android devices.

### Phase 22: Multi-KVK Agronomic Field Trials & Longitudinal Clinical Validation

• Strategic Need: Formal academic and clinical validation against certified agricultural scientists.
• Execution: Double-blind pilot trials across 4 Maharashtra Krishi Vigyan Kendras (KVK Kolhapur, KVK Pune/Baramati, KVK Nashik, KVK Ahmednagar). Compares 1,000 live field leaf scans diagnosed concurrently by PikVaidya AI and certified pathologists.
• Acceptance Criteria: Diagnostic concordance $\ge 95.0\%$; zero critical chemical safety infractions reported by KVK officers.

### Phase 23: Production Kubernetes Cloud Cluster & Multi-Region CI/CD

• Strategic Need: High-availability infrastructure capable of supporting millions of Indian smallholders.
• Architecture: Multi-stage Docker containers orchestrated via Kubernetes (K8s) with Horizontal Pod Autoscaling (HPA), PostgreSQL 16 with read-replicas and PgBouncer pooling, Redis 7 caching cluster, Prometheus/Grafana, and automated GitHub Actions CI/CD.
• Acceptance Criteria: 99.95% system uptime; p95 API response $< 350$ ms under sustained load of 5,000 concurrent requests/sec.

## 5. Phased Implementation Timeline & Milestone Sprints

The proposed execution trajectory spans 4 focused implementation sprints following the completed baseline:

| Milestone Period | Active Phases | Key Engineering Deliverables | Target Outcome |
|---|---|---|---|
| **Current Baseline** | Phases 1 – 16 | Vision models, Grad-CAM, Platt calibration, Indic RAG, IoT gateway, offline PWA, conversational chat, 19-step E2E test. | **240/240 tests passing (100%). Robust, verified micro-monolith ready.** |
| **Sprint 1**<br>**Weeks 1 – 3** | Phase 17 | Cotton & Soybean dataset ingestion, parent-grouped splits, MobileNetV3 fine-tuning, calibration, CIBRC seeds. | **5-Crop Coverage: Sugarcane, Tomato, Onion, Cotton, Soybean verified.** |
| **Sprint 2**<br>**Weeks 4 – 6** | Phase 18 & 19 | Whisper/FastSpeech2 Indic streaming voice engine; standalone MQTT broker background daemon. | **Hands-free trilingual voice UX; real-time sensor node streaming.** |
| **Sprint 3**<br>**Weeks 7 – 9** | Phase 20 & 21 | Sentinel-2 MSI 10m satellite vegetation index pipeline; Capacitor Android native packaging. | **Macro satellite verification; Google Play Store distributable release APK.** |
| **Sprint 4**<br>**Weeks 10 – 14** | Phase 22 & 23 | Multi-KVK clinical agronomic trials (Kolhapur, Pune, Nashik, Ahmednagar); Kubernetes cluster scaling & CI/CD. | **100% Commercial Field Readiness, certified KVK concordance, cloud scaling.** |

## 6. Formal Verification & Engineering Sign-Off

This document constitutes the official engineering specification for the complete PikVaidya AI platform. All claims regarding Phases 1 through 16 are verified by executable source code, trained neural model weights, and passing automated test suites in the repository.

| Audit Role | Officer / Sentinel | Verification Artifact | Status |
|---|---|---|---|
| Chief AI Architect | Senior Software Architect | Checkpoints, Calibration ($T^*$), Grad-CAM Hooks | **APPROVED** |
| Lead Systems Engineer | Sentinel-BackendGateway | FastAPI Core, SQLAlchemy Models, Alembic Migrations | **APPROVED** |
| Lead IoT Specialist | Sentinel-EdgeIoT | LoRaWAN Ingestion, 24h Summary, RAG In-Situ Fusion | **APPROVED** |
| Principal QA Specialist | Sentinel-DevOpsQA | Automated Pytest Matrix (240 / 240 Passed, 100%) | **VERIFIED** |

### Academic References & Standards:

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